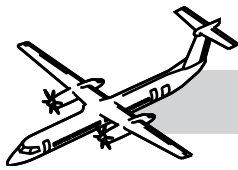


# **CHAPTER 74 IGNITION**

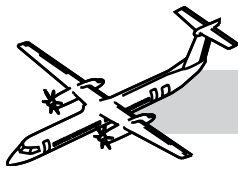
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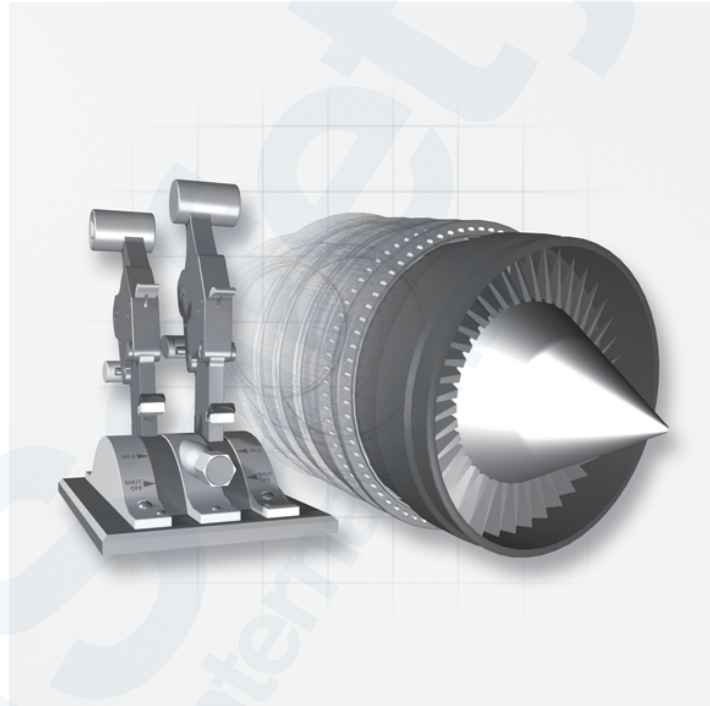


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# CHAPTER 74 IGNITION



## 74-00-00 INTRODUCTION

The ignition system supplies the electrical energy to ignite the fuel/air mixture during engine start-up. It also provides automatic in flight re-light capability in the event of a flame out or a STALL/SURGE.

## GENERAL

The system function is performed by the ignition exciters, cables and plugs. The ignition system includes the following systems:

- Power Supply
- Ignition Distribution
- Ignition Selection.

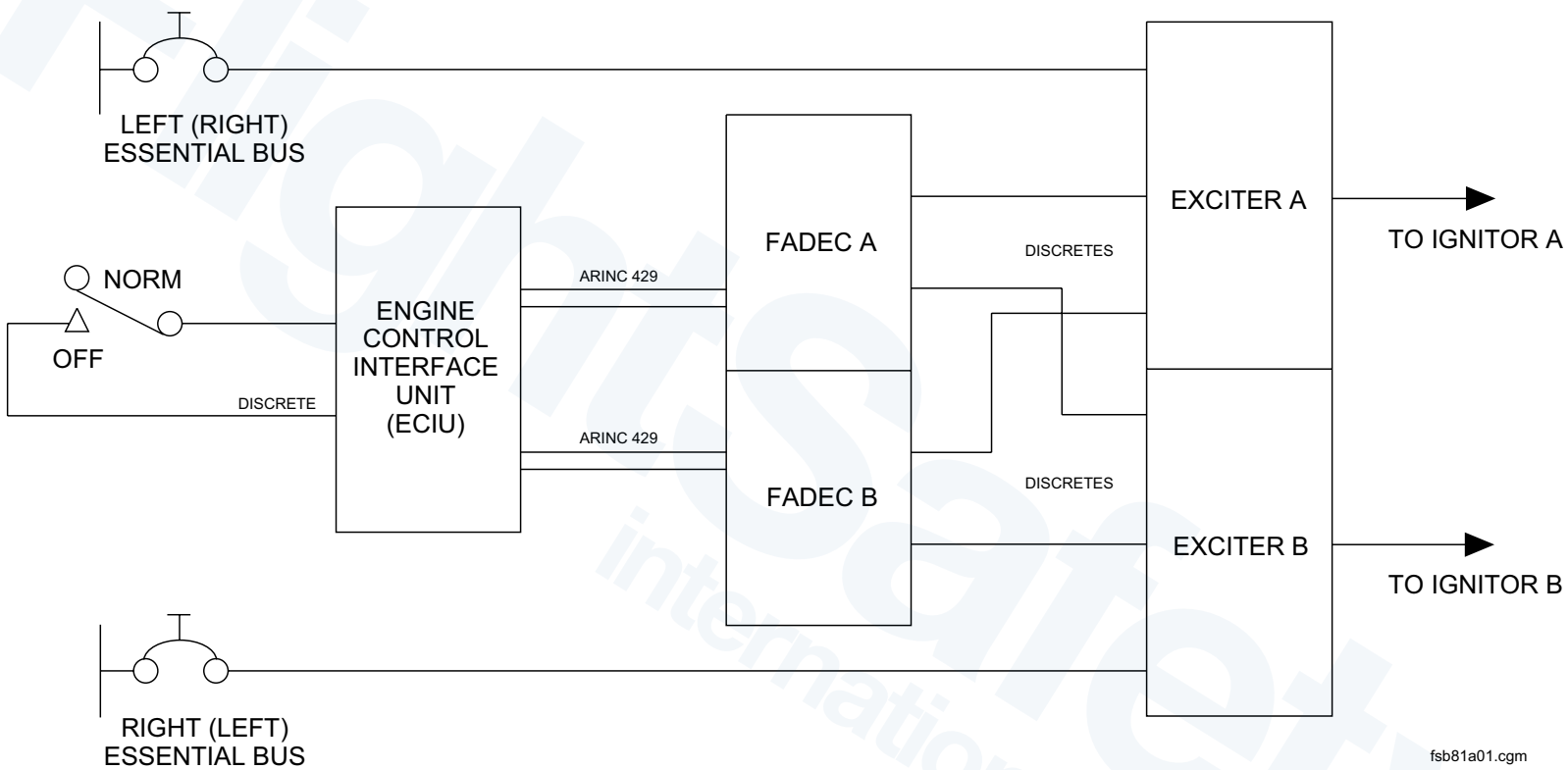
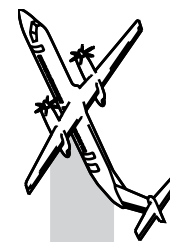
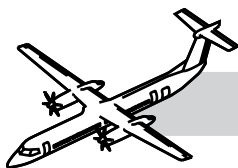


Figure 74-1. Ignition System Block Diagram



## SYSTEM DESCRIPTION

## NOTES

Refer to Figure 74-1. Ignition System Block Diagram.

The PW150A ignition system has:

- Ignition exciter
- FADEC
- ECIU
- Ignition cables
- Ignition plugs.

The ignition exciter power is supplied from 28 VDC through a single electrical connector for both channels.

It outputs high voltage to the igniter plugs when commanded “ON”. The ignition exciter is connected to the igniter plugs by a pair of braided cables.

The cables are mounted with quick release connectors to facilitate removal and installation. The igniter plugs are in the gas generator case.

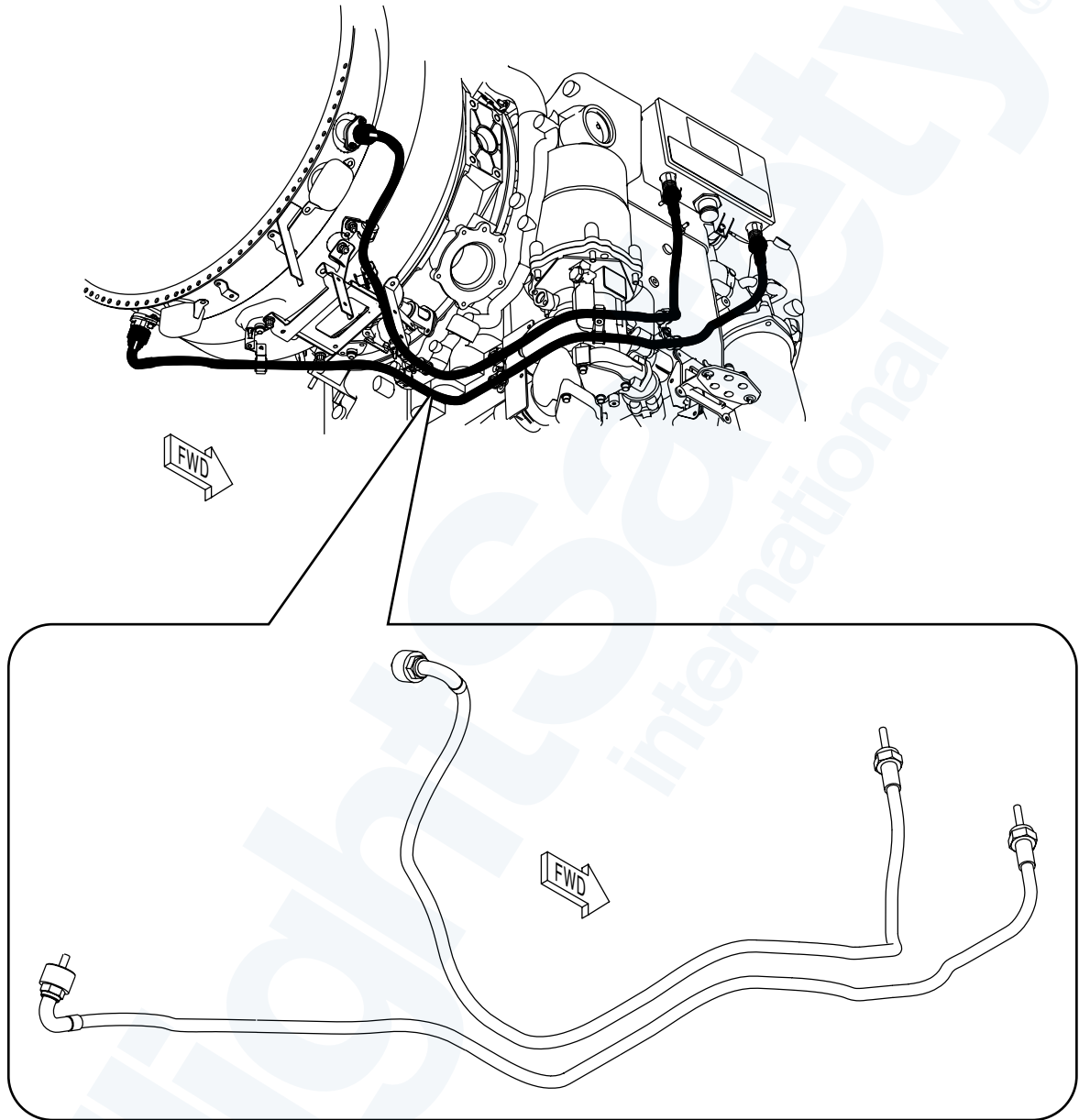
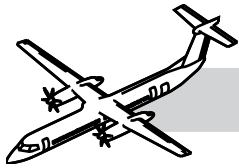
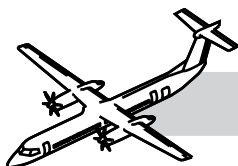


Figure 74-2. Ignition Distribution Cables



## IGNITION EXCITER AND IGNITION CABLES

Refer to:

- Figure 74-2. Ignition Distribution Cables.
- Figure 74-3. Ignition Exciter.

The ignition exciter transforms DC voltage input into a pulsed high voltage output to provide the energy to the ignitor plugs.

When power is supplied to the ignition exciter, the energy level is increased to a high enough level to cause the ignition plugs to spark. The ignition exciter is a sealed unit and must be returned to an overhaul facility for repair.

The cables are on the right side of the engine.

### WARNING

HIGH ENERGY IN THE IGNITION SYSTEM CAN BE FATAL. OBSERVE ALL SAFETY PRECAUTIONS IN THE AMM CHAPTER 74 BEFORE WORKING ON THE SYSTEM.



Figure 74-3. Ignition Exciter

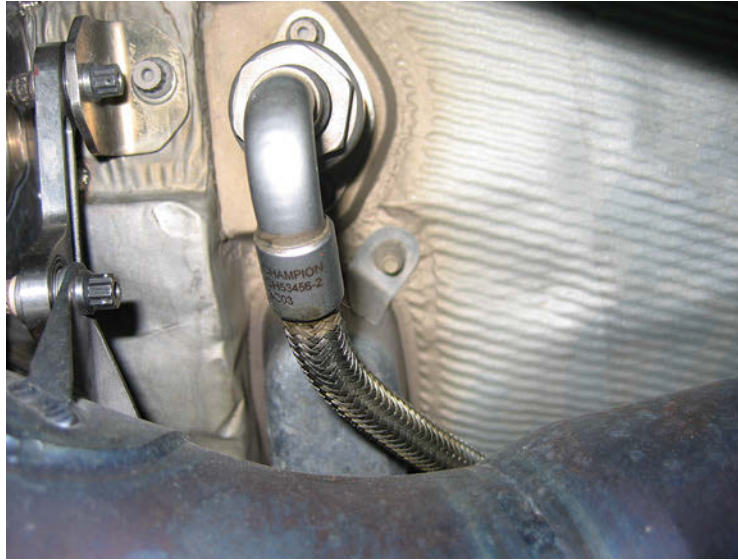
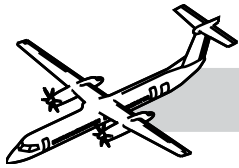


Figure 74-4. Ignition plugs

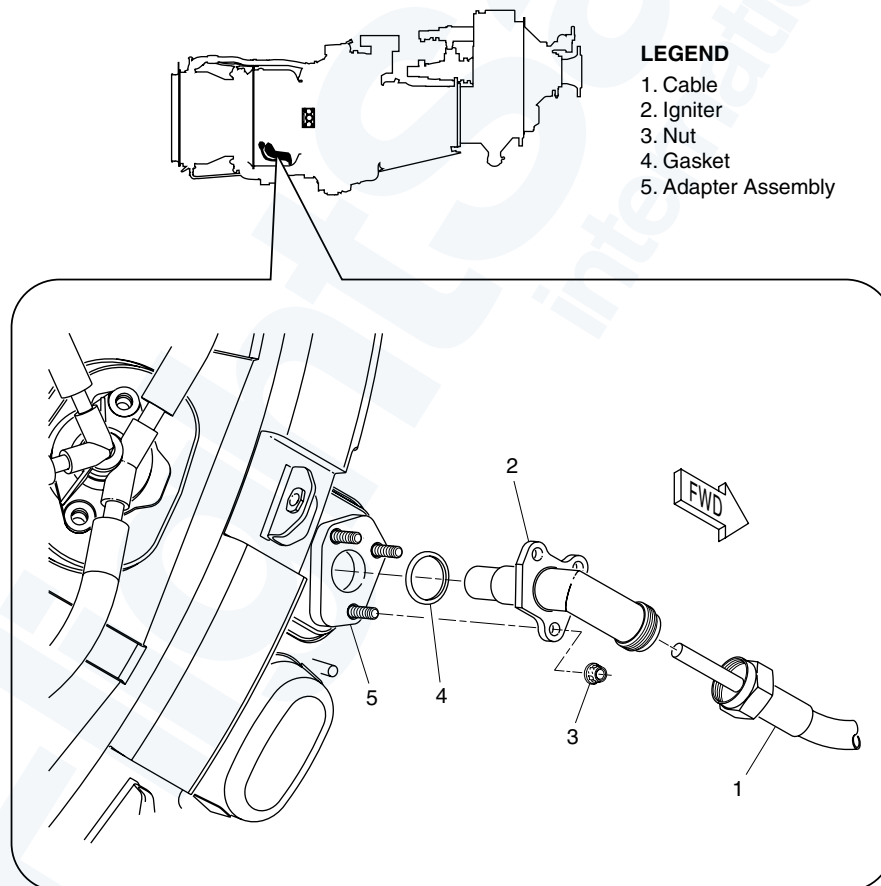
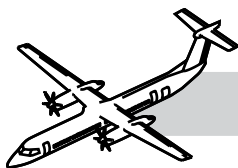


Figure 74-5. Igniter Plugs - Post SB35140





## IGNITION PLUGS

## NOTES

Refer to Figure 74-4. Ignition plugs.

There are two ignition plugs. They are installed in the gas generator case at the 4 and 7 o'clock positions. The ignition plugs are connected to the Ignition Exciter Box by the Ignition Distribution Cables.

### Training Information Points

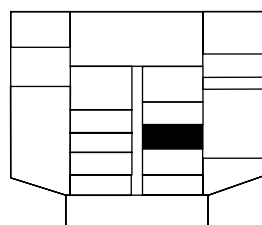
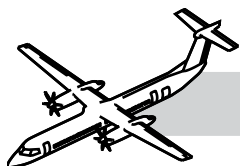
The ignition plugs are the drop in type. Be careful when you disconnect the ignition cables and attaching hardware because the plugs can fall out.

If you drop a plug you must replace it with a new one because it may be damaged internally.

## IGNITION PLUG REMOVAL

*The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.*

- Obey all the electrical/electronic safety precautions
- Obey electrostatic discharge safety precautions
- De-energize electrical system
- Open and tag CBs in the sequence shown in the task
- Disconnect the ignition cable from the plug
- Remove the nuts from the adapter
- Remove the igniter plug from the adapter
- Obey all warnings and cautions mentioned in the task sheet during the execution of this task.



OVERHEAD CONSOLE

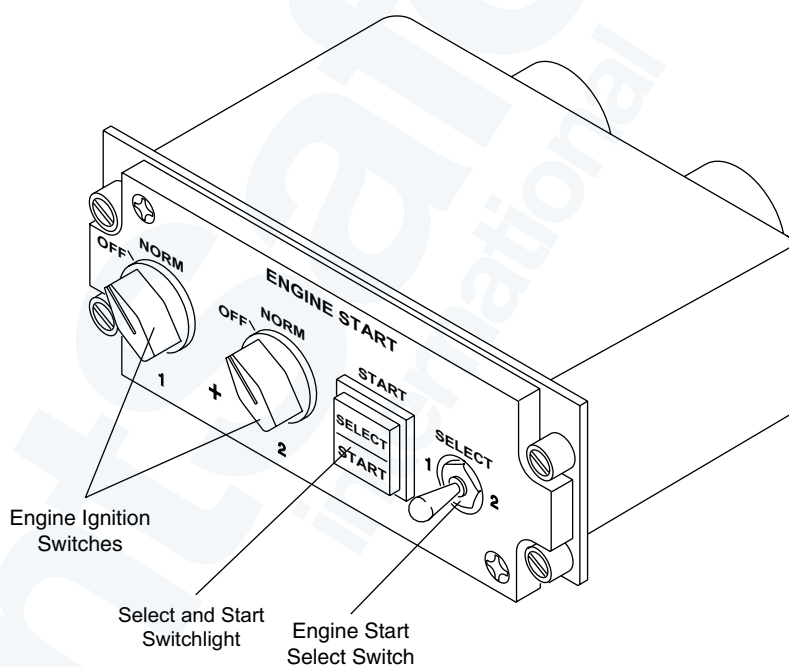
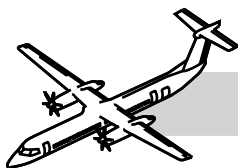


Figure 74-6. Engine Start Panel



## ENGINE START PANEL

## NOTES

Refer to Figure 74-6. Engine Start Panel.

The Engine Start Panel is in the flight compartment on the overhead console. It has switches that are used for engine ignition selection. This panel is labeled ENGINE START and has the following switches:

- Ignition selection switches
- Engine starter select switch
- Engine start push-switch.

## IGNITION SELECTION SWITCHES

The ignition selection switches allow the flightcrew to select the ignition mode. There are two ignition modes; “NORMAL” and “OFF”. When the switches are in the “NORMAL” position, the FADEC activates ignition during engine starts (on the ground or in-flight). When the switches are in the “OFF” position, the FADEC disables ignition.

### Engine Starter Select Switch

The Engine Starter Select Switch is used to select the left or right engine starter.

The start select switch will be described in detail in ATA 80.

### Engine Start Push-Switch

This switch signals the related FADEC to activate the selected engine's starter.

The start select switch will be described in detail in ATA 80.

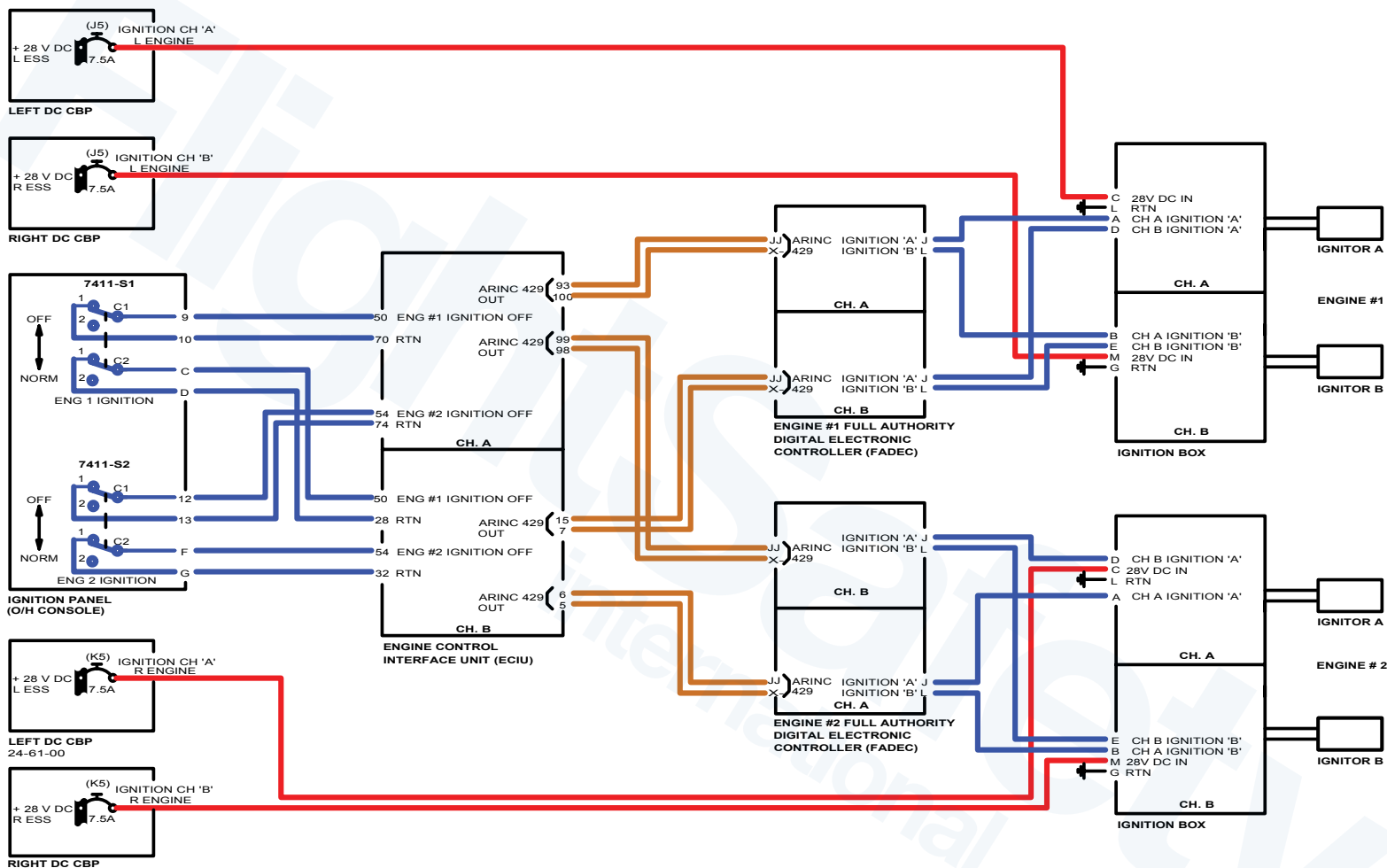
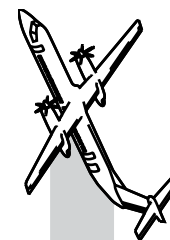
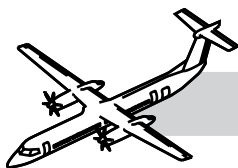


Figure 74-7. Engine Ignition System



## OPERATION

### NORMAL SYSTEM OPERATION

Refer to Figure 74-7. Engine Ignition System..

Each exciter has a channel “A” and a channel “B”. The left essential bus supplies electrical power to channel “B”. The right essential bus supplies electrical power to channel “A”. With the BATTERY MASTER selected ON the exciters, are powered, and wait for a signal from the FADEC to operate.

The switch on the engine start control panel when in the OFF position provides continuity “OFF” discrete to channel A and B of the ECIU. In the Norm Position it provides a failsafe “ON” discrete to channel A and B of the ECIU. The ECIU changes the analog signal to a digital signal and each channel sends a signal to channel A and B of each FADEC on an ARINC 429 bus.

FADEC will default to normal if it loses communication with the ECIU.

When FADEC detects Nh is greater than 8%, it sends a discrete to the exciter and commands one ignitor to operate. Each channel of FADEC can operate either ignitor plug. On subsequent starts the other ignitor is used., this is to minimize ignitor wear.

After the start, Nh greater than 50%, FADEC will select the ignitor “Off”.

### ABNORMAL OPERATION

FADEC will command both ignitors to operate if:

- FADEC detects no light off after 8 seconds from selected Fuel On
- FADEC detects there is a flameout when airborne
- FADEC detects that there is a surge or compressor stall.

If an ignitor does not operate, FADEC will log a fault count of 3 to the ignitor. If the ignitor reaches a count of 9 then a fault code is logged for the faulty ignitor.

### IGNITION FUNCTIONAL TEST

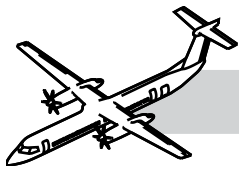
For the right, or left engine, do the functional test of the igniter plugs as followings:

1. Set the PLA to FLIGHT IDLE
2. Set the CLA to FUEL OFF
3. Set the Ignition to NORMAL
4. Check the igniters as follows:
  - A. Select MAINT DISC on the ENGINE MAINTENANCE section of the Central Maintenance Panel.
  - B. Select MCR on the Engine Control Panel.
  - C. Check if the igniter sparks

#### NOTE

When the test is in progress, the FADEC will alternately energize the two igniters. During the test, the ITT gauge will indicate 0°C (32°F) when igniter A (7 o'clock) fires and 1792°C when igniter B fires. If you listen near the engine combustion chamber, you can hear the igniter plug when it sparks.

- D. If you do not hear a spark, do the Fault Isolation Procedure (refer to FIM 73-20-00-810-822)
- E. Set the Ignition to OFF
- F. Select MAINT DISC to OFF on the ENGINE MAINTENANCE section.



## 74-00-00 APPENDIX

### MAINTENANCE CONSIDERATION

#### Safety Precautions

##### **WARNING**

DO NOT TOUCH THE IGNITION EXCITER LESS THAN SIX MINUTES AFTER THE IGNITION SYSTEM STOPS. THE REMAINING VOLTAGE IN THE IGNITION EXCITER CAN BE VERY HIGH. THIS HIGH VOLTAGE CAN CAUSE INJURY TO PERSONS.

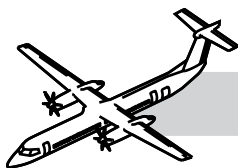
DO NOT TOUCH THE ELECTRICAL CONNECTOR CONTACT WITH BARE HANDS. THE MATERIAL OF THE ELECTRICAL CONNECTOR CONTACT IS DANGEROUS. YOU CAN GET SKIN IRRITATION IF YOU TOUCH THE CONTACT WITH YOUR BARE HANDS.

##### **CAUTION**

DO NOT LET THE CABLE BRAIDING OR FERRULES TURN WHEN YOU TURN THE COUPLING NUTS. IF YOU LET THE CABLE BRAIDING OR FERRULES TURN, DAMAGE CAN OCCUR TO THE CABLE.

MAKE SURE THAT THE IGNITERS DO NOT FALL ON THE GROUND WHEN YOU DISCONNECT THE CABLES FROM THE ADAPTERS. INTERNAL DAMAGE CAN

OCCUR IF YOU LET AN IGNITER PLUG DROP ON THE GROUND. A TEST OF THE PLUG CAN NOT ALWAYS FIND THIS DAMAGE. IF AN IGNITER PLUG FALLS ON THE GROUND, REPLACE IT.



## **74-00-00 SPECIAL TOOL & TEST EQUIPMENT**

- Commercially Available Ohmmeter
- PWC55732 Cap

## **74-00-00 MAINTENANCE PRACTICES**

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 74-11-01-000-801: Removal of the Ignition Exciter.
- AMM 74-11-01-400-801: Installation of the Ignition Exciter.
- AMM 74-21-01-000-801: Removal of the Ignition Plug.
- AMM 74-21-01-400-801: Installation of the Ignition Plug.
- AMM 74-21-06-000-801: Removal of the Ignition Cable.
- AMM 74-21-06-400-801: Installation of the Ignition Cable.
- AMM 74-21-01-720-801: Functional Test of the Igniter Plugs.